Vihaan Manchanda
Duke University · Explainable AI

prompting-explained

Which Words in Your Prompt Actually Matter?

A teaching tool that shows you the impact of each phrase in your prompt on the AI's answer.

SCAN TO TRY THE APP
github.com/Vihaan8/prompting-explained

01 Executive summary

People in almost every field now use AI tools, but there is no tool that teaches prompt-writing literacy. AI developers learn what works through practice. Most users never do. In one study, non-experts wrote prompts the way they would instruct a person, and struggled to fix them when the answer went wrong (Zamfirescu-Pereira et al., 2023). Small wording changes matter too. Changing only a prompt's formatting moved one model's accuracy by up to 76 points (Sclar et al., 2024). A badly worded prompt wastes tokens and context, and gets a worse answer.

I built **prompting-explained** to teach how each word in a prompt contributed to the AI's answer. It removes one phrase at a time, reruns the prompt, and highlights the phrases that changed the result. On 240 test prompts across 8 everyday tasks, the phrases it ranked highest mattered clearly more than random ones, and its top pick repeated 78% of the time.

The goal is a training tool companies load with their own tasks, so staff learn what works on their real work.

02 The problem

People learn to prompt by trial and error. Chat tools return an answer but never show which words the model acted on, so nobody can tell which of their habits help. For a company rolling out AI, that costs three things:

- **Worse answers.** Missing or counterproductive instructions go unnoticed.
- **Wasted tokens.** In my 240 test prompts, 45% of phrases asked for something the AI would have done anyway.
- **Hidden risk.** An answer can look fine while quietly breaking an instruction, such as keeping private details out, and nothing warns the user.

Prompting guides offer general tips. They don't show people what their own prompt did.

03 Who it's for

Everyday users. People who just want the right answer. They see which of their words helped and which didn't, without needing to know how AI works.

Employees at companies adopting AI. Staff who are still learning to use AI practice on their own company's tasks, documents and rules.

Governance and compliance teams. Test whether a safety instruction, such as "leave out customer details", actually changes the answer before rolling it out.

THE QUESTION

Can removing one phrase at a time and rerunning the prompt show which words changed the AI's answer, in a way that is **faithful** and **stable**?

04 How it works

1 Split the prompt into phrases
An AI splits the prompt into short phrases. Code keeps each phrase exactly as typed, typos included.

↓

2 Remove one phrase at a time and rerun
The full prompt runs 3 times, and each version with one phrase removed runs twice (occlusion*). Phrases that do nothing alone are also removed together once, in case they cover for each other.

↓

3 Score every answer against the task's goal
Each task turns its goal into a short checklist that code grades from 0 to 1. Every version of the prompt is graded the same way, so what matters is how the score changes.

↓

4 Measure each phrase's effect
A phrase's effect is how far the score moves when it is removed. If the score drops, the phrase helped. If it rises, the phrase hurt.

↓

5 Decide what each phrase did
Answers vary from run to run, so a change only counts if it is bigger than that phrase's noise floor*.

Is the change bigger than the noise floor?

↓ YES	↓ NO
Score dropped: helped	Did a second AI see the instruction carried out only with the phrase in?
Score rose: hurt	Yes: followed / No: no effect

*** Occlusion (leave-one-out):** hide one part of the input and measure the change, giving each phrase a counterfactual. LIME does this with single words.
*** Noise floor:** two standard errors of the run-to-run variation. Smaller changes stay grey.

05 Worked example

Task: write a prompt that gets the AI to explain inflation to your 10-year-old cousin. One user wrote this:

> Hi there! Explain inflation to my 10-year-old cousin. Use exactly 3 bullet points. Mention a candy bar in one of them. Use proper technical terms.

Green · "Explain inflation": without it, the AI explained how the body uses food for energy.

Blue · "Mention a candy bar": the AI did it, but the score was the same either way.

Red · "Use proper technical terms": the AI used words like "Consumer Price Index", too hard for a 10-year-old. Without it, the score rose from 0.40 to 0.78.

Grey · "Hi there!": a greeting gives the AI nothing to act on.

On a fresh rerun, 5 of 6 verdicts repeated. The "without it" sentences are written by an AI and labelled as descriptions; every number is computed by code.

HOW THE COLORS ANSWER THE THREE PROBLEMS

- Helped the answer → **Worse answers**
- Hurt the answer → **Worse answers**
- No real difference → **Wasted tokens**
- Followed, score unchanged → **Hidden risk**

06 What I found

DOES THE TOOL WORK?

It points at the right phrases. Removing the two phrases it ranked highest dropped the score by 0.30, against 0.18 for two random phrases (79 prompts, p = 0.0002). Its explanations are faithful.

It gives the same answer twice. A full rerun picked the same top phrase 78% of the time. Nearly tripling the reruns only raised agreement from 83% to 91%, so its explanations are stable.

It beat my own guesses. Before running anything, I wrote down what I expected each phrase to do. I matched the tool on 76% of phrases in clean prompts but only 52% in messy ones. Reading the answers, about 35 of the 41 disagreements were my mistakes.

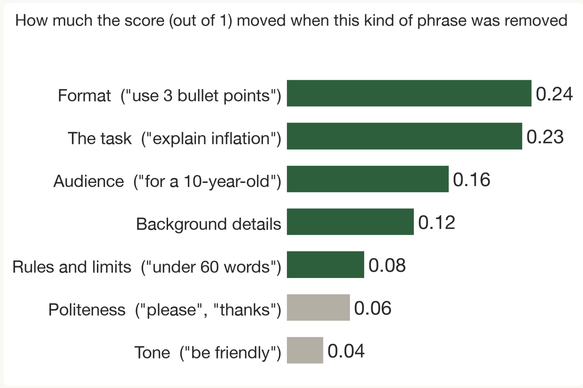
WHAT IT SHOWED ABOUT PROMPTS

One model and my 240 test prompts, so these are observations, not rules.

Format and the task moved the score most, partly because some of my checklists reward a format the task never stated, like exactly 3 bullets. Politeness and tone barely moved it (chart, right).

Length limits matter for cost. In 60 prompts with a word limit, removing it made answers longer in 49, by 45% at the median.

Some instructions backfire. "dont guess" made the AI speculate *more*, in all 5 runs.



Average change in score when each kind of phrase was removed, across 741 phrases in 240 test prompts.

07 What I can't claim yet

- **That people prompt better after using it.** There is no study with real users yet.
- **That the results hold beyond my setup.** One model (Claude Haiku 5.5), and test prompts written by an AI imitating users, plus 32 messy ones I wrote.
- **Much about tasks where a document does the work.** There, removing two phrases moved the score by less than 0.07.
- **That my checklists are neutral.** Some reward details the task never stated, like exactly 3 bullets, so a prompt can lose points for not guessing them.
- **That every check is right.** The AI that checks instructions passed 93 answers that broke a word limit.
- **That deletion tells the whole story.** Phrases are tested alone and once as a group, not with better wording or in every combination, as Shapley values would.

08 What's next

- **For companies.** Employees learn what works on their own tasks and documents, and compliance teams check a safety instruction before it goes into shared prompts.
- **Lower cost (stretch goal).** Cache measured phrases and train a small local model, so an app can flag a missing length limit or wasted words before a prompt is sent, with no extra AI call.
- **First, a study with real users.** Do people write better prompts after using it?

In these tests, the tool reliably pointed at the words that drove an answer, and it beat my own guesses. Whether it makes people write better prompts is the next thing to test.

REFERENCES Zamfirescu-Pereira et al. (2023), Why Johnny Can't Prompt, CHI · Sclar et al. (2024), prompt formatting sensitivity, ICLR · Zeiler & Fergus (2014), occlusion · Ribeiro et al. (2016), LIME · Lundberg & Lee (2017), SHAP · Petsiuk et al. (2018), deletion test · DeYoung et al. (2020), faithfulness · Alvarez-Melis & Jaakkola (2018), stability · Turpin et al. (2023), unfaithful model explanations